STATISTICAL ENVIRONMENTS + WORKED EXAMPLE / ABOUT 10 MINUTES

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The peak that grows while gRMS stays still

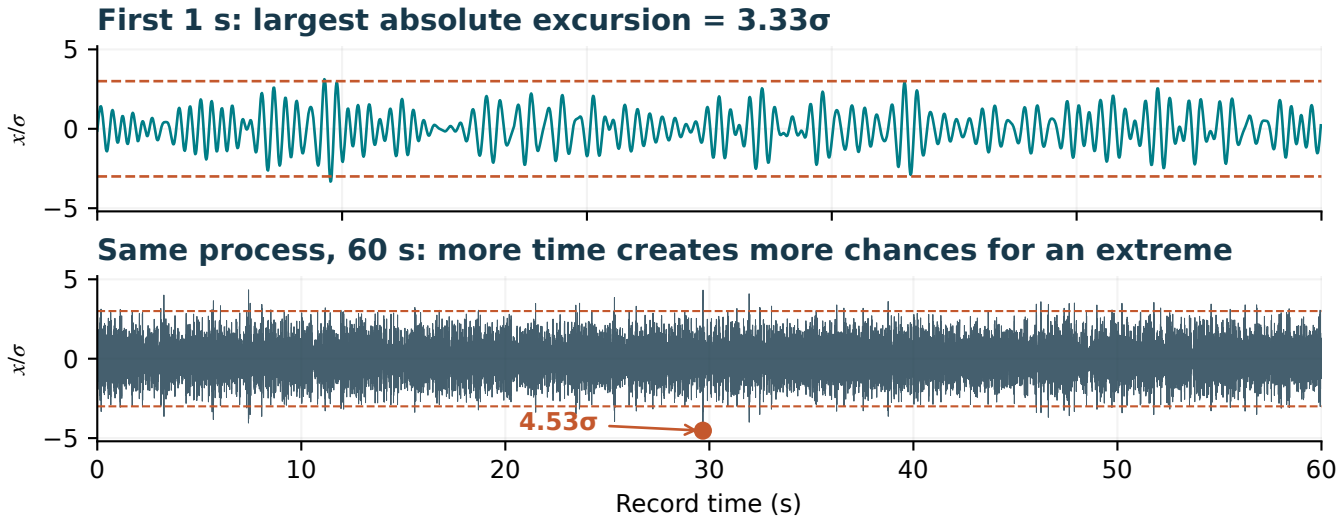


Figure 1. Original conceptual simulation: stationary, Gaussian, 75–125 Hz band-limited noise normalized to 1 RMS. The individual extrema are one realization, not measured data or a design criterion.

The shaker is holding the commanded PSD. Overall response has settled at 1 g_{RMS}. Nothing about the structure or the control spectrum changes—yet the largest number in the recorder keeps creeping upward. At one second the record may look like a tidy “three-sigma” process. By a minute, it has simply had many more chances to produce an uncommon excursion.

That is the puzzle for today: RMS is a stable second-moment statistic; the largest peak is an extreme-value statistic tied to record length and spectral content. A multiplier such as 3×RMS can be a program limit-load convention, but it is not a time-independent ceiling on a stationary Gaussian response. NASA-STD-5002B says random-vibration limit loads are *typically* taken as 3σ or computed to the required enclosure level, then immediately ties the multiplier to the assumed distribution and required enclosure/confidence. [5]

The objective is to connect PSD moments, crossing rate, duration and peak probability; work a narrowband example; and turn “use 3σ” into a reviewable statement with scope, probability and limitations.

01 / THE RMS CAN CONVERGE WHILE THE MAXIMUM DOES NOT

Two statistics, two different questions

For a zero-mean stationary response $x(t)$ with one-sided PSD $S_x(f)$, the variance is the area under the PSD:

$$\sigma_x^2 = m_0 = \int_0^\infty S_x(f)\,\mathrm{d}f$$

Once the PSD estimate and operating state are stable, that RMS can settle. The maximum of a finite record, $M_T = max\ |x(t)|$, asks a different question: how far into the probability tail did this particular record wander? Longer duration creates more opportunities. Faster fluctuations create more opportunities per second.

The distinction begins before any formula. An arbitrary time sample from a Gaussian response is normally distributed. A *local maximum* is a sample conditioned on zero slope and negative curvature, so its distribution is different. The single largest local maximum adds another selection step across the entire record. Applying the normal cumulative distribution directly to that last quantity silently discards the time structure that the PSD contains.

The frequency-domain bridge is the spectral-moment crossing rate. Define angular-frequency moments

$$m_j = \int_0^\infty (2\pi f)^j\, S_x(f)\,\mathrm{d}f$$

For a differentiable stationary Gaussian process, Rice's result gives the expected positive-slope zero crossings per second as

$$\nu_0^+ = \frac{1}{2\pi}\sqrt{\frac{m_2}{m_0}}$$

For a narrowband response centered near f_c, this is approximately f_c. Over duration T, use $N \approx \nu_0^+ T$ as an opportunity count for cycles. Bendat's NASA report develops the crossing-rate and peak-probability results and emphasizes their assumptions; it also shows that narrowband Gaussian positive peaks follow a Rayleigh distribution. [1]

This is why two responses with the same RMS can have different likely maxima. A 20 Hz resonant response and a 500 Hz resonant response observed for the same minute do not present the same number of excursions, even before nonstationarity or non-Gaussian behavior enters the problem.

02 / A NARROWBAND CALCULATION YOU CAN DO ON A REVIEW SLIDE

What “five sigma in 100,000 cycles” means

For a narrowband Gaussian process, the positive peak envelope R is Rayleigh. Normalizing by RMS, the probability that one positive envelope peak exceeds $k\sigma$ is

$$P(R > k\sigma) = \exp\left(-\frac{k^2}{2}\right)$$

Treat N positive peaks and N negative peaks as approximately independent opportunities. Then the probability that the largest absolute peak in the record does not exceed $k\sigma$ is

$$P(M_T \le k\sigma) \approx \left[1 - \exp\left(-\frac{k^2}{2}\right)\right]^{2N}$$

This independence approximation is useful for intuition and a narrowband screening calculation; it is not the general wideband peak formula. Solve for the p-nonexceedance peak factor:

$$k_p = \sqrt{-2\ln\left(1 - p^{1/(2N)}\right)}$$

Worked example—illustrative, not measured: a lightly damped response is dominated by a 100 Hz mode during a 1,000 s stationary exposure. Taking $N \approx 100{,}000$ cycles gives

$$P(M_T > 5\sigma) \approx 1 - (1 - e^{-12.5})^{200000} = 0.525$$

So a five-sigma absolute extreme is slightly more likely than not under this particular model and counting convention. The ECSSMET 2016 paper reported the same practical message—extrema above 3σ were systematic in recorded random tests and the probability of a 5σ extreme exceeded 50% for 100,000 cycles—while also showing that input clipping does not guarantee clipped structural responses. [3]

The factor is not a deterministic prediction of the next test. It is a quantile of a stated stochastic model. A different duration, bandwidth, tail model, or definition of “cycle” changes it.

03 / DURATION AND BANDWIDTH MOVE THE ANSWER

One RMS, several credible maxima

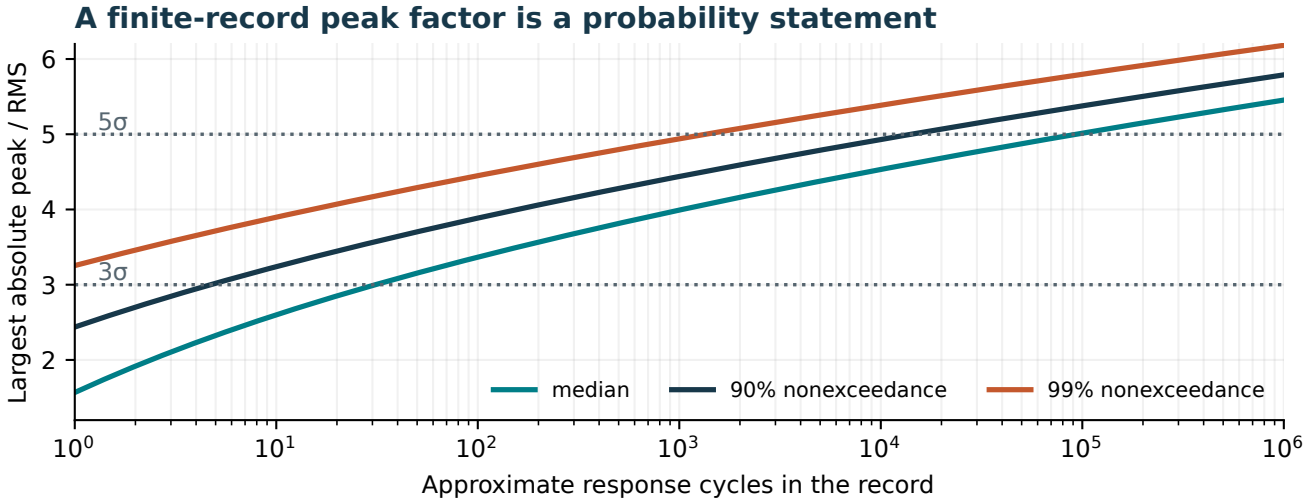


Figure 2. Original calculation using the narrowband Rayleigh-peak approximation and two signed extrema per cycle. Curves are finite-record quantiles, not standards or acceptance limits.

100 Hz response	Cycles, N	Median |peak|	P(|peak| > 5σ)
10 s	1,000	3.99σ	0.7%
60 s	6,000	4.42σ	4.4%
600 s	60,000	4.91σ	36.1%
1,000 s	100,000	5.01σ	52.5%

Table 1. Same 1-RMS narrowband process, different record lengths. The median maximum rises only logarithmically, but it does keep rising. “3σ contains 99.73% of instantaneous normal samples” does not mean a 3σ absolute maximum has 99.73% nonexceedance over a long record. Samples and record maxima are different random variables.

Bandwidth enters more subtly than “center frequency.” For a general Gaussian response, spectral moments determine zero-crossing and peak rates, while a bandwidth parameter influences the distribution of local maxima. A broad response contains many differently shaped excursions and correlated samples; counting FFT lines or digitizer samples as independent trials is wrong. Use established wideband extreme-value methods or direct, statistically adequate time-history ensembles when the narrowband assumption is weak. Bendat treats the distinction explicitly. [1]

04 / PUT THE PEAK FACTOR IN ITS PROPER JOB

A design-review workflow

Start with the decision. RMS response, fatigue damage, clearance, yield, contact, latch release and electronics upset are not interchangeable endpoints. RMS or PSD is natural for stationary energy and linear response. A maximum-load or clearance check needs an extreme-value model. Fatigue needs the cycle-amplitude distribution and duration, not merely the single largest peak.

State the stochastic model. Record whether the response is assumed stationary, Gaussian and narrowband; define the PSD convention and spectral moments; state duration and whether the factor refers to a positive peak, negative peak or absolute maximum. A bare “3σ” hides every one of these choices.

Separate environment statistics from response statistics. A Gaussian base acceleration through a linear system remains Gaussian, but its bandwidth and crossing rate change. A nonlinear joint, clipping, intermittent forcing, impacts, limiters or response-dependent stiffness can change the distribution and tail. Rizzi’s NASA study notes that crest factor depends on record length and uses kurtosis to describe distribution-wide non-Gaussianity; one extreme alone is a noisy descriptor. [4]

Use the duration that belongs to the question. Qualification exposure, flight segment, combined mission time and repeated missions can have different durations and stationarity assumptions. If the environment is time-varying, divide it into defensible stationary segments or use a nonstationary method. Barnoski’s NASA study showed finite-time peak response depends on duration and, for short records, damping and build-up; stationary results can bound some shaped-noise cases, but only within the studied oscillator assumptions. [2]

Validate the tail. Compare histogram, skewness, kurtosis, crest factor versus subrecord duration, peak rate and level crossings. Inspect stationarity and clipping. Use multiple realizations or confidence bounds. If a test controller clips drive peaks, verify response channels independently—the dynamics can regenerate extrema beyond the drive clipping level. [3]

Keep requirements scoped. NASA-STD-7001C, dated July 14, 2026, addresses acoustic and random-vibration environments and test levels for covered spaceflight payload hardware; it is not a universal rule for launch vehicles, aircraft, sounding-rocket payloads or ground equipment. Program requirements and technical authority remain controlling. [6]

05 / TAKEAWAYS, REFLECTION AND SOURCES

Three takeaways

1. A finite-record peak factor needs duration. RMS can remain fixed while the observed maximum grows because each additional cycle offers another tail opportunity.

2. Bandwidth controls opportunity rate and peak shape. Estimate crossings from PSD moments; do not treat sampled points or FFT lines as independent peaks. Use narrowband formulas only when the response is narrowband enough to justify them.

3. "3σ" is a convention unless probability is attached. State the distribution, signed/absolute definition, exposure duration, enclosure probability and scope. For nonlinear or nonstationary response, validate the tail directly.

A short reflection

A component has a 12 g_{RMS} narrowband response near 250 Hz for 120 s. Is 36 g a defensible maximum solely because it is 3σ?

Answer: No. The record contains roughly 30,000 response cycles; under the illustrative narrowband Gaussian model, the median absolute maximum is about $4.77\sigma = 57$ g. The correct design value still depends on the required nonexceedance probability, actual bandwidth, stationarity, tail behavior and governing program criteria. The calculation is a screening estimate, not a qualification limit.

References

[1] J. S. Bendat, Probability Functions for Random Responses: Prediction of Peaks, Fatigue Damage, and Catastrophic Failures, NASA CR-33, 1964. See Sections 2–5 and 8.

[2] R. L. Barnoski, The Mechanical Response of a Linear Mechanical Oscillator to Stationary and Nonstationary Random Excitation, NASA CR-340, December 1965.

[3] B. Laine et al., "Dynamic Tests: What's Behind the Curve?", ECSSMET 2016. Test observations and analytical discussion; conference paper, not a standard.

[4] S. A. Rizzi et al., "On the Response of a Nonlinear Structure to High Kurtosis Non-Gaussian Random Loadings," 2011. Public NASA technical paper.

[5] NASA-STD-5002B, Load Analyses of Spacecraft and Payloads, March 10, 2026. Appendix C.4; apply only when invoked and within scope.

[6] NASA-STD-7001C, Payload Vibroacoustic Test Criteria, July 14, 2026. Scope excludes launch vehicles, sounding-rocket payloads, aircraft, balloons and GSE.

Figures 1–2 and Table 1 are original illustrative calculations. No measured data, vehicle telemetry or program requirement is represented. This run could not verify or select GPT-6 Astra.